Abuthahir B

Data Analyst | Data Scientist | ML Engineer

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SUMMARY

Self-motivated data professional with 5+ years of experience supporting AI-based products and contributing to AIML projects. Hands-on experience in RAG architecture, semantic retrieval, FAISS-based vector search, and NLP-based systems using LLMs and embedding models. Experienced in Computer Vision image annotation (bounding box, polyline, polygon) for AI model training.Completed B.E in Computer Science Engineering (2025) while working full-time. Skilled in Python, SQL, Machine Learning, NLP, and Deep Learning. Built end-to-end ML and NLP projects with Streamlit deployments. Certified by GUVI, IIT Madras Research Park. Seeking ML/NLP Engineering Intern role to contribute to LLM-based systems and document understanding pipelines.

PROFESSIONAL EXPERIENCE

Junior Team Leader – AI Product Operations 02/2021 – Present
Mahendra Next Wealth IT India Pvt Ltd Salem,Tamilnadu

- Led a 60+ member team supporting an AI-powered retail store product
- Contributed to 3 POC projects and 1 live AIML-based exception handling project (24/7 production)
- Performed bounding box, polyline, and polygon image annotation for Computer Vision AI model training
- Prepared 50+ weekly MIS reports; tracked KPIs and identified operational trends

EDUCATION

B.E Computer Science and engineering 2022 – 2025
Annaporana Engineering College Salem,Tamilnadu
(Pursued while working full-time)

Diploma – Mechanical Engineering 06/2018 – 04/2020
Mahendra Polytechnic College Salem,Tamilnadu

SKILLS

Python, SQL, MS Excel, Pandas, Power BI, Streamlit, Data Analysis, Machine Learning, NLP, Deep Learning, LLM, RAG, FAISS, Semantic Analysis, Vector Embeddings, HuggingFace

LANGUAGES

- English
- Tamil
- Hindi

CERTIFICATES

Data Science Program – GUVI, IIT Madras Research Park — Modules: Python | SQL | Pandas | NumPy | EDA | Machine Learning | NLP | Deep Learning | Data Visualization

PROJECTS

Telecom AI Copilot

- Developed a telecom AI assistant using RAG architecture for document -grounded customer support response generation
- Implemented semantic retrieval using HuggingFace pretrained transformer model (all-MiniLM-L6-v2) for converting telecom queries and documents into vector embeddings.
- Integrated FAISS for efficient vector similarity search and retrieval of relevant telecom policy and pricing information.
- Designed evaluation pipeline to measure retrieval accuracy and response quality of the LLM-based system
- Built document understanding pipeline to extract and synthesize context from telecom policy documents

IMDb Movie Recommendation System (NLP)

- Developed an **IMDb Movie Recommendation System** using **Python, Selenium, NLP, TF-IDF, and Cosine Similarity** to recommend movies based on storyline similarity by scraping and analyzing IMDb 2024 movie data.
- Built an interactive **Streamlit application** that processes user-input storylines and provides the top 5 similar movie recommendations using NLP-based text preprocessing, vectorization, and similarity scoring techniques.

Swiggy Restaurant Recommendation System

- Developed a **Restaurant Recommendation System** using **Python, Pandas, Streamlit, and Machine Learning** techniques to provide personalized restaurant suggestions based on city, cuisine, rating, and cost preferences.
- Implemented **data preprocessing, One-Hot Encoding, and similarity/clustering-based recommendation algorithms** (K-Means/Cosine Similarity), along with an interactive Streamlit dashboard for real-time restaurant recommendations and user-friendly data visualization.

NutriClass: Food Classification Using Nutritional Data

- Developed **NutriClass**, a multi-class food classification system using **Python, Pandas, Scikit-learn, and Machine Learning** models to predict exact food categories based on nutritional attributes such as calories, protein, carbohydrates, fats, and sugar.
- Performed end-to-end **data preprocessing, feature engineering, model training, and evaluation** by implementing algorithms like Logistic Regression, Random Forest, SVM, XGBoost, and KNN, achieving optimized classification performance using Accuracy, Precision, Recall, F1-score, and Confusion Matrix analysis.

Data-Driven Stock Analysis

- Developed a **Data-Driven Stock Analysis Dashboard** using **Python, Pandas, SQL, Streamlit, and Power BI** to analyze and visualize Nifty 50 stock performance, including volatility, cumulative returns, sector-wise trends, and stock correlations.
- Built interactive dashboards and automated data pipelines to process YAML-based stock datasets into structured CSV/SQL formats, enabling real-time insights on top gainers/losers, market trends, and investment performance metrics.

Client Query Management System

- Developed a **Client Query Management System** using **Python, MySQL, Pandas, and Streamlit** to manage real-time client support queries with secure login authentication, query tracking, and status management.
- Built an interactive **Streamlit dashboard** for support teams to monitor, filter, and close queries, while analyzing query trends and resolution performance using SQL queries and datetime-based tracking.